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A feature selection method with feature ranking using genetic programming

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ABSTRACT

Feature selection is a data processing method which aims to select effective feature subsets from original features. Feature selection based on evolutionary computation (EC) algorithms can often achieve better classification performance because of their global search ability. However, feature selection methods using EC cannot get rid of invalid features effectively. A small number of invalid features still exist till the termination of the algorithms. In this paper, a feature selection method using genetic programming (GP) combined with feature ranking (FRFS) is proposed. It is assumed that the more the original features appear in the GP individuals' terminal nodes, the more valuable these features are. To further decrease the number of selected features, FRFS using a multi-criteria fitness function which is named as MFRFS is investigated. Experiments on 15 datasets show that FRFS can obtain higher classification performance with smaller number of features compared with the feature selection method without feature ranking. MFRFS further reduces the number of features while maintaining the classification performance compared with FRFS. Comparisons with five benchmark techniques show that MFRFS can achieve better classification performance.

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KEYWORDS

feature selection; feature ranking; genetic programming; multi-criteria

1. Introduction

Feature selection (Sreeja, 2019; Too & Abdullah, 2020) is to select effective feature subsets from high-dimensional original features, which is one of the key issues for machine learning. High-quality features play a key role in building an efficient model, and irrelevant or redundant features may cause difficulties (Xue et al., 2013). Therefore, feature selection methods have been extensively applied to solve practical classification tasks (Espejo et al., 2010; Liang et al., 2017; Loughran et al., 2017; Mei et al., 2017; Patel & Upadhyay, 2020). Recently, Gaining Sharing Knowledge based Algorithm (GSK) (Mohamed et al., 2020) was proposed for feature selection and achieved good performance (Agrawal, Ganesh, & Mohamed, 2021a, 2021b; Agrawal, Ganesh, Oliva, et al., 2022). Evolutionary computation (EC) algorithms (Al-Sahaf et al., 2019; Hancer, 2019) have been widely used for feature selection due to their global search ability (Ma & Gao, 2020b; Xue et al., 2014). EC algorithms are population based,

which need to initialise individuals randomly. Sometimes, the genetic operators of genetic algorithm (GA) and genetic programming (GP), or the updating strategy of particle swarm optimisation (PSO) (Nagra et al., 2020; Too et al., 2021), are not sufficient to delete invalid features. So, a small number of invalid features still exist till the termination of the algorithms.

GP can be used for feature selection due to its global search ability and is proved to achieve good classification performance. However, with the evolution of GP, a large number of individuals with the same best fitness are generated. GP only outputs the best individual, which may lose some good individuals. Moreover, the generated features in the output individual may still contain redundant features.

Feature ranking (Ahmed et al., 2014; Friedlander et al., 2011; Neshatian, 2010) is the ranking of original features based on specific evaluation criteria, which is usually a step of feature selection. It is employed to find out which features or feature sets are more important. So, this paper proposes a GP-based feature selection method combined with feature ranking (FRFS), which considers the large number of best individuals generated during the evolution of GP instead of only one best individual. A correlation-based evaluation criterion is used as the fitness function. It is assumed that the more the original features appear in the GP individuals' terminal nodes, the more valuable these features are. Therefore, the individuals with the best fitness are stored. The occurrence times of features appearing in the individuals' terminal nodes are counted and the top β features are selected as the candidate feature subset. To decrease the number of selected features while maintaining the classification performance, another feature selection method that combines FRFS with a multi-criteria fitness function is investigated. The motivation of this algorithm is to set higher fitness values for the individuals with the same correlation value in the case of smaller feature number.

The overall objective of this paper is to propose a feature selection algorithm that combines feature ranking with a multi-criteria fitness function and verify the improvement of the algorithm by feature ranking and the dimensionality reduction effect of multi-criteria fitness function. To achieve the overall objective, the following three objectives will be investigated.

Objective 1. Propose a feature selection method combined with feature ranking using GP (FRFS) and verify whether it can achieve better classification performance with smaller number of features than using feature selection only (FS).

Objective 2. Investigate a feature selection method that combines FRFS with a multi-criteria fitness function which is named as MFRFS, and verify whether the multi-criteria fitness function can reduce the number of features of FRFS while maintaining the classification performance.

Objective 3. Investigate whether our proposed MFRFS can achieve better classification performance than five benchmark techniques, including ReliefF feature selection method (Kira & Rendell, 1992), linear forward feature selection method (LFS) (Gutlein et al., 2009) and three GSK related methods (Agrawal, Ganesh, & Mohamed, 2021b; Agrawal, Ganesh, Oliva, et al., 2022).

The rest of this paper is arranged as follows. Section 2 outlines the background information of this paper. Section 3 describes in detail the three GP-based feature selection methods. Section 4 shows the experimental design. Section 5 presents the experimental results and discussion. Section 6 is the conclusion and future work.

2. Related works

2.1. Genetic programming (GP)

GP is one of the evolutionary computation algorithms (Koza, 1992; Koza et al., 1999). It is very similar to genetic algorithm (GA). The main difference between GP and GA is the representation of individuals. Due to GP's flexible representation methods, GP can be used to construct high-level features (Ma & Gao, 2020a, 2020b), construct classifiers (Muni et al., 2006; Neshatian, 2010; Neshatian & Zhang, 2011) and solve practical industrial problems (Bi et al., 2021a, 2021b; Peng et al., 2020). The commonly used representation method of GP is based on tree structure. The terminal nodes (constants and variables) are randomly selected from a terminal set, and the function (mathematical and logical operators) is randomly selected from a function set to constitute internal nodes. GP randomly initialises the first generation of individuals and evaluates each individual using a fitness function. Genetic operators including selection, crossover and mutation are then performed to produce the next generation's population. This step is iterated until the termination criterion is reached. Then GP outputs the optimal individual. For feature selection, the terminal nodes contain the features selected by GP.

2.2. Correlation

The correlation-based feature evaluation criterion is proposed by Hall (1999), which takes into account both the correlation between features and classes and the correlation between features. This evaluation criterion is proved to be effective to select low redundancy and high discrimination feature subsets (Hall & Smith, 1999; Ma & Gao, 2020b), and is adopted as the evaluation criterion in this paper. The following is the formula for calculating the correlation of a feature subset.

$$\text{Correlation}_S = \frac{k\overline{C}_{fc}}{\sqrt{k + k(k-1)\overline{C}_{ff}}} \quad (1)$$

where Correlation_S is the correlation value of a set S containing k features, $\overline{C}_{fc}$ denotes the average correlation between each feature and class, and $\overline{C}_{ff}$ denotes the average correlation between each feature pair. They are calculated by formula (2). $H(X)$ and $H(Y)$ are the information entropy of X and Y respectively, $H(X|Y)$ is the conditional entropy.

$$C_{XY} = 2.0 \times \left[\frac{H(X) - H(X|Y)}{H(X) + H(Y)} \right] \quad (2)$$

2.3. Feature selection methods based on GP

Due to GP's flexible representation methods, GP can be used for feature selection. Recent studies have focused on feature selection methods using evolutionary algorithms (Canuto & Nascimento, 2012; Papa et al., 2017; Ribeiro et al., 2012; Tan et al., 2008) due to their global search ability. According to whether the classification algorithm is involved in the fitness function, feature selection is divided into filter-based (Davis et al., 2006; Lin et al., 2008;

Neshatian & Zhang, 2009b, 2012; Purohit et al., 2010; Ribeiro et al., 2012) and wrapper-based (Hunt et al., 2012). The filter-based feature selection methods use information measures such as mutual information (Vergara & Estévez, 2014), information gain (Neshatian & Zhang, 2012), consistency and correlation (Hall, 1999; Neshatian & Zhang, 2009c) as the evaluation criterion, and need less running time than wrapper-based methods. Moreover, the models produced by filter-based methods are more general. So, filter-based feature selection is investigated further in this paper. The fundamental research goal of feature selection is to find the feature subset with the best classification performance and fewer features.

Lin et al. (2008) constructed a classifier using layered genetic programming, which had the characteristic of feature selection and feature extraction. Neshatian and Zhang (2012) combined information entropy and conditional entropy to evaluate the correlation between feature sets and classes for feature selection.

Purohit et al. (2010) proposed a GP classifier construction method with feature selection and investigated a new crossover operator to discover the best crossover site. Davis et al. (2006) proposed a two-stage feature selection method. In the first stage, a feature subset is selected using GP and GP classifiers are evolved using the selected features. Neshatian and Zhang (2009b) proposed a filter-based multiple-objective feature selection method for binary classification task.

GP has flexible representation ability. In general, the GP-based feature selection method takes original features as the terminal set and terminal nodes as the selected features. However, some researchers have researched other representations. Hunt et al. (2012) developed GP-based hyper-heuristics for adding and removing features. Each individual generated by GP is a series of operations on the feature set. Ribeiro et al. (2012) used four feature selection criteria (information gain, chi-square, correlation and odds ratio) as the terminal set of GP and three set operations (intersection, union and difference) as the function set. Viegas et al. (2018) proposed a GP-based feature selection method for skewed dataset and validated the method on four high-dimensional datasets. Papa et al. (2017) used binary strings as individual's terminal nodes, where 1 means selecting a certain feature and 0 means without selecting. AND, OR and XOR are regarded as non-terminal nodes. The output of the individual is also a binary string, which contains the selected features.

2.4. Feature ranking methods based on GP

The goal of feature ranking is to show the importance of the original features under certain evaluation criteria and is often used as a basis of feature selection.

Neshatian (2010) proposed a feature ranking method based on GP, which used GP to evolve many weak classifiers and preserved the optimal weak classifiers. The score of each feature is proportional to the fitness value of the weak classifier that contains it. This method needs to run a large number of GPs, which is time-consuming.

Friedlander et al. (2011) proposed a feature ranking method based on weight vectors. GP updates the weights for each feature at the end of each generation of evolution. After 10 generations of evolution, terminal nodes are no longer randomly selected during the generation of new subtrees in mutation operation, but are selected according to different probabilities based on feature weights.

Ahmed et al. (2014) proposed a two-stage GP-based classification method which first ranked the original features then constructed classifiers. Neshatian and Zhang (2009a) constructed GP programs to measure the goodness of feature subsets using a virtual program structure and an evaluation function.

At present, there have been many researches on GP-based feature selection methods, but few research focus on the large amount of evolutionary information generated during the evolution process. And some researchers have begun to use GP to rank the features (Friedlander et al., 2011; Neshatian, 2010), which indicates that the frequency of features appearing in terminal nodes can show the importance of features. Based on this motivation, some issues should be investigated on whether feature ranking can help quantify the importance of features and improve the classification performance, and whether a multi-criteria fitness function can help to further explore this area select more effective features.

3. Method

In this paper, our proposed methods are based on standard GP, and correlation is chosen as the fitness function. The fundamental purpose of this paper is to investigate whether the feature selection method based on correlation evaluation criteria using GP can effectively get rid of invalid features, and whether feature ranking and a multi-criteria fitness function can improve the classification performance. We have different variants of algorithms to verify the research objectives. FS denotes the feature selection method using GP. FRFS denotes the feature selection method combined with feature ranking. MFRFS denotes the algorithm that combines FRFS with a multi-criteria fitness function. The details of the algorithms are described below.

3.1. Feature selection using GP (FS)

This algorithm employs GP as the search algorithm and correlation as the evaluation criterion. Because the smaller the fitness of GP, the better. The fitness function of FS is shown in Formula (3).

$$\text{fit}_{\text{FS}} = (1 - \text{Correlation}) \quad (3)$$

where Correlation is the correlation value of the selected features and classes.

The output of FS is an individual that contains the optimal feature subset. The terminal nodes are the selected features. The parameter settings of GP are shown in Section 4.1.

3.2. FS combined with feature ranking (FRFS)

FS only considers the best individual generated by GP and does not make full use of the information during GP's evolutionary process. With the convergence of GP, there are a large number of individuals with the same best fitness. Our next step, feature ranking, is based on the assumption that the more frequent features occur in these individuals, the more valuable they are. All the individuals with the same best fitness during the evolution process after the j th generation are preserved. The occurrence of features in these individuals' terminal nodes are counted, and the features are ranked according to their occurrence. Our

Algorithm 1: Pseudo-Code of FRFS algorithm

Input: D : a training set
Output: D_s : the feature set selected by FRFS

- 1 Random initialisation of the first generation population of GP
- 2 **while** *Maximum number of generations is not reached* **do**
- 3 **for** $i = 1; n \leq s; i++$ **do**
- 4 Calculate fit_i of the individual i using Correlation
- 5 **if** $currentgeneration > 30$ **then**
- 6 **if** $fit_i < fit_b$ **then**
- 7 $fit_b = fit_i$ // fit_b is the current optimal fitness value
- 8 Empty storage pool and store individual i to storage pool
- 9 **end**
- 10 **else if** $fit_i == fit_b$ **then**
- 11 Store individual i to storage pool
- 12 **end**
- 13 **end**
- 14 **end**
- 15 Perform evolutionary operations: reproduction, crossover and mutation
- 16 **end**
- 17 Get the number of features k of the optimal individual
- 18 Count occurrences of features contained by all individuals in the storage pool and get the sequence R in descending order
- 19 **for** $j = \frac{k}{3}; j \leq k; j++$ **do**
- 20 The first j features are extracted from the sequence R and the classification accuracy acc_j is obtained on the training set by the classifier
- 21 **if** $acc_j > acc_b$ **then**
- 22 $acc_b = acc_j$
- 23 $f_{best} = j$
- 24 // acc_b is the largest classification accuracy value, and f_{best} is the number of features matching with acc_b
- 25 **end**
- 26 Return the first f_{best} features of sequence R
- 27 **end**

experiments found that GP starts to converge after 30 iterations, so the parameter j is set to 30.

Suppose the number of features selected by FS is k . Our goal is to select the top f_{best} features among top k features according to ranking. The Pseudo-Code of FRFS is shown in Algorithm 1. To determine how many features selected from k features are the best, first $f_{best} = \frac{k}{n}$ and the f_{best} features are tested on the classification algorithms. Then the rest $k - \frac{k}{n}$ features are added into f_{best} feature subset and tested one by one according to their ranking until the classification accuracy begins to decline. Then, the top f_{best} features are the selected features. In the paper, the parameter n is set to 3, that is, top one-third of k

features are selected into f_{best} feature subset. We started with top $\frac{k}{n}$ features instead of the top feature is to improve the search efficiency. We set the value of n to 3 based on our experiments.

3.3. FRFS combined with a multi-criteria fitness function (MFRFS)

In the process of GP evolution, we found that two individuals with the same fitness value are not necessarily the same individuals. To give priority to individuals with the same fitness value but less features, we use the multi-criteria fitness function as Formula 4. The method FRFS using the multi-criteria fitness function is named as MFRFS.

$$\text{fit}_{\text{MFRFS}} = (1 - \text{Correlation}) + \alpha f_{\text{num}} \quad (4)$$

where f_{num} is the number of features in an individual and α is the penalty coefficient between Correlation and f_{num} . The parameter α is discussed in Section 5.4.

4. Experimental design

4.1. Datasets and parameter settings

To verify the effectiveness of our proposed feature selection methods, 15 datasets are collected from UCI machine learning repository (Dheeru & Karra Taniskidou, 2017). The details of the datasets are shown in Table 1. In the table, #features indicates the number of original features, #instances indicates the number of instances and #classes indicates the number of class labels. The datasets have different number of classes and features, and are representative classification problems.

A random split of 7:3 is used and 70% of the dataset is used as the training set and 30% as the testing set (Ma & Gao, 2020b; Xue et al., 2013). The training set is used to select and rank the original features. The testing set is used to evaluate the classification performance of selected and ranked features by 10-fold cross-validation. To avoid stochastic characteristics of GP, 30 experiments are done independently (Ma & Teng, 2019). The average of classification accuracy and number of features of 30 different experiments are obtained.

Table 1. Description of the fifteen datasets.

Dataset	#features	#instances	#classes
arrhythmia	262	452	13
audiology	69	226	24
climate-simulation-crashes	20	540	2
hepatitis	19	155	2
house-vote	16	435	2
Lymphoma	4026	51	3
planning-relax	12	182	2
robot-failure-lp4	90	117	3
robot-failure-lp5	90	164	5
seismic-bumps	18	2584	2
spambase	57	4601	2
spect-heart	22	267	2
statlog-heart	13	270	2
statlog-image-segmentation	19	2310	7
thoracic-surgery	16	470	2

Table 2. Parameter setting of GP.

Parameters	Parameter value
Function set	+, −, *, % (protected division)
Terminal set	Original features F0
Population size	500
Number of generations	50
Initialisation	Ramped half-and-half
Mutation method	Random subtree creation
Selection method	Tournament method
Maximum tree depth	17
Mutation probability	10%
Cross-over probability	90%
Reproduction probability	10%

K-nearest Neighbours (KNN, $K = 5$), C4.5 decision tree and Naive Bayes (NB) are used to evaluate the performance of the proposed methods. The ECJ platform (Luke, 2017) is used to run GP. The parameter settings of GP are shown in Table 2, which are the commonly used parameters of GP (Ma & Gao, 2020b; Ma & Teng, 2019). The function set is composed of four arithmetic operators, the terminal set is the original features. The population size is set to 500, initial GP experimental results indicate that this population size can evolve good solutions. The generations are set to 50 and the maximum tree depth is 17 to restrict irrelevant and redundant features can also reduce the depth of GP programs. Mutation probability and crossover probability are 10% and 90% respectively, which balance exploration and exploitation during the evolution. The parameter α in Formula 3 is set as 0.01, which is discussed in Section 5.4.

4.2. Benchmark techniques

To verify the effectiveness of the proposed MFRFS methods, we use five existing feature selection methods as benchmarks to compare, including ReliefF feature selection method (Kira & Rendell, 1992), linear forward feature selection method (LFS) (Gutlein et al., 2009) and three GSK-related methods (Agrawal, Ganesh, & Mohamed, 2021b; Agrawal, Ganesh, Oliva, et al., 2022).

4.2.1. ReliefF

ReliefF (Kira & Rendell, 1992) is an extension of Relief algorithm. Relief can only be used to solve binary classification problems. Both Relief and ReliefF are feature weighting algorithms, which assign different weights to features according to the correlation between each feature and class, and features with a weight less than a certain threshold will be deleted. The weight of a feature is proportional to its classification ability. ReliefF can be applied to solve multi-class classification problems. First, one sample is randomly selected from the dataset, and then k nearest neighbour samples are selected from all classes. Then, the weight of all features is calculated, and the operation is repeated n times. Finally, the final weight of each feature is obtained and the features are ranked according to their weights. To facilitate comparisons with the proposed MFRFS, the same number of features as MFRFS is selected on all datasets by ReliefF.

4.2.2. LFS

LFS (Gutlein et al., 2009) starts from an empty set and adds features one by one. It restricts the number of features in each step of the forward selection. When the evaluation criteria show that its performance remains unchanged or decreases after adding a feature, LFS will be terminated. LFS is proved to be faster than standard forward selection and can obtain good results (Gutlein et al., 2009). To compare with the proposed MFRFS method fairly, Correlation is also selected as the evaluation criteria.

4.2.3. GSK related methods

Gaining-sharing knowledge-based optimisation algorithm (GSK) (Agrawal, Ganesh, & Mohamed, 2021b) is a human-related metaheuristic algorithm. The basic principle of GSK is to simulate the process of human gaining and sharing knowledge in the life span. Recently, GSK has been used for feature selection and proved to be effective, so three GSK-related methods including BGSK (Agrawal, Ganesh, & Mohamed, 2021b), pBGSK (Agrawal, Ganesh, & Mohamed, 2021b) and PbGSK-V₄ (Agrawal, Ganesh, Oliva, et al., 2022) were selected as benchmarks for comparison.

5. Experimental results and conclusion

To verify the effectiveness of our proposed methods, four experiments are done. (1) FRFS is compared with two baselines, i.e. original features (F0) and FS, to verify whether FRFS can get higher classification accuracy with fewer features. (2) MFRFS is compared with FRFS to verify whether the multi-criteria fitness function used by MFRFS can maintain the classification performance and reduce the number of features. (3) MFRFS is compared with five benchmarks, i.e. ReliefF, LFS, BGSK, pBGSK and PbGSK-V₄, to verify whether MFRFS can achieve better classification performance in terms of accuracy and number of features. (4) The parameter α used in MFRFS is discussed to determine an appropriate parameter setting.

To justify the significance of various feature selection methods, the performance differences between paired feature selection methods are shown on the boxplots in Figures 1 and 2. The post-hoc analysis for Friedman test (Hollander & Wolfe, 1999) abbreviated as

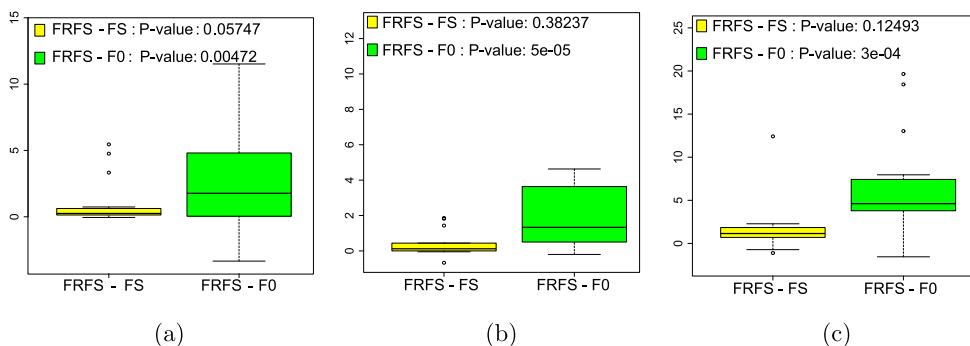


Figure 1. The boxplots of performance differences between FRFS and two baselines marked by Post-Friedman-test: (a) KNN, (b) C4.5, (c) NB.

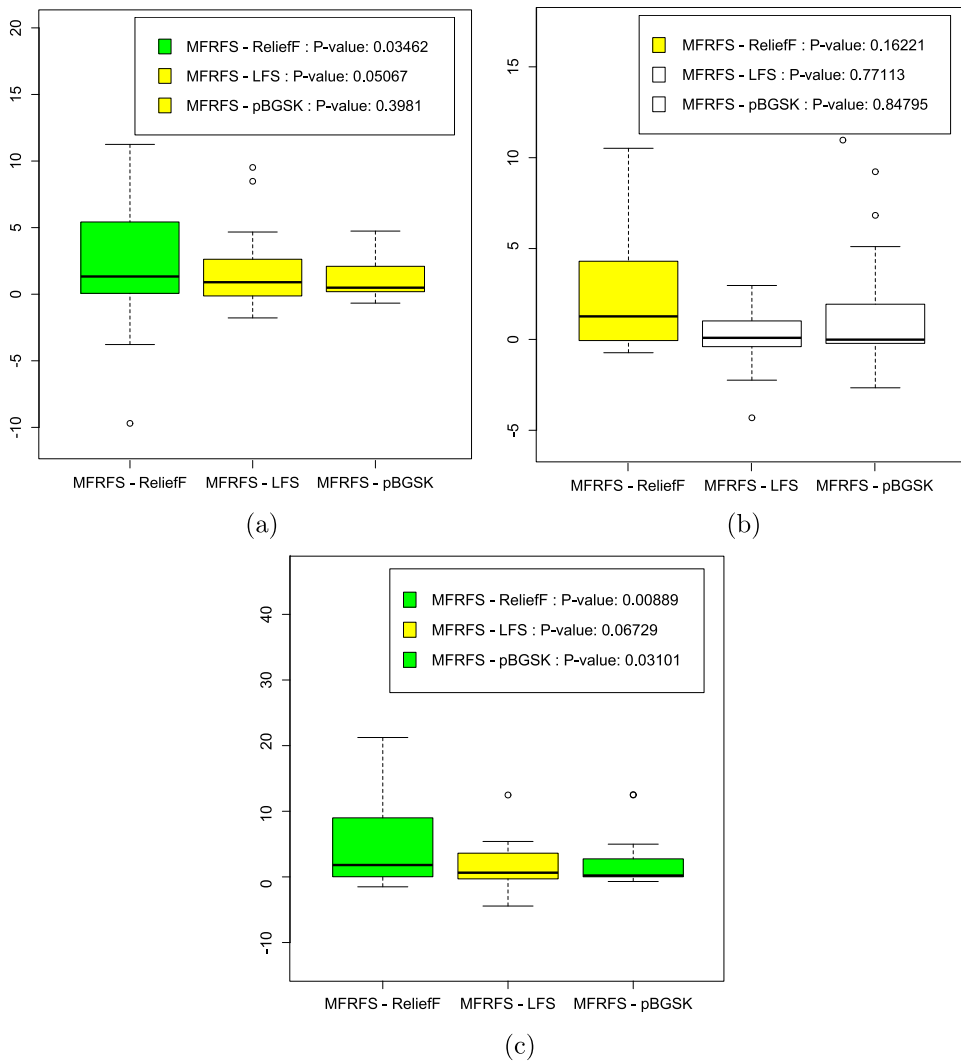


Figure 2. The box plots of performance differences between MFRFS and three benchmarks marked by Post-Friedman test: (a) KNN, (b) C4.5, (c) NB.

Post-Friedman test using different colours is also marked on the boxplots. If the Post-Friedman test achieves $p\text{-value} \leq 0.05$ or $0.05 < p\text{-value} \leq 0.4$, the box plots are marked green or yellow respectively, which shows that the performance differences between paired feature processing methods are significantly different or borderline significantly different. Figures 1 and 2 show whether one of the paired methods is significantly better than the other.

5.1. Comparison between FRFS and F0, FS

In this section, the experiments between FRFS and F0, FS were done and the experimental results are shown in Table 3. In the table, A-KNN, A-C4.5 and A-NB represent the average

Table 3. The experimental results of FRFS, F0 and FS.

Dataset	Methods	A-KNN	#F-KNN	A-C4.5	#F-C4.5	A-NB	#F-NB
arrhythmia	F0	61.33	262	65.67	262	61.67	262
	FS	64.17	14.6	65.92	14.6	65.92	14.6
	FRFS	64.42	10.5	66.17	10.5	66.75	11.6
audiology	F0	60.89	69	78.67	69	61.56	69
	FS	66.89	5.8	80	5.8	78.67	5.8
	FRFS	70.22	4.9	80	4.9	80	4.9
climate-simulation-crashes	F0	92.01	20	88.93	20	88.87	20
	FS	91.19	2.9	91.76	2.9	86.86	2.9
	FRFS	91.51	2.4	92.2	2	89.12	2.3
hepatitis	F0	85.56	19	80.67	19	85.33	19
	FS	82	6.9	81.56	6.9	82.67	6.9
	FRFS	82.22	4.6	82	3.3	83.78	5.3
house-vote	F0	91.59	16	95.13	16	89.97	16
	FS	95.74	1	95.63	1	95.63	1
	FRFS	95.74	1	95.63	1	95.63	1
Lymphoma	F0	80	4026	86	4026	80	4026
	FS	80	19.7	90.67	19.7	82	19.7
	FRFS	80	11.1	90	9.8	84	10.8
planning-relax	F0	66.85	12	66.3	12	60.93	12
	FS	67.78	4.5	69.07	4.5	67.59	4.5
	FRFS	68.52	2.7	70.93	2.1	68.89	2.4
robot-failure-lp4	F0	50.3	90	66.36	90	80	90
	FS	56.36	27.6	75.15	27.6	80.61	27.6
	FRFS	61.82	11.7	76.97	11.9	83.03	16.2
robot-failure-lp5	F0	41.43	90	44.29	90	59.05	90
	FS	42.14	26.6	45	26.6	61.9	26.6
	FRFS	46.9	15.4	46.43	19.7	62.86	21.8
seismic-bumps	F0	92.97	18	93	18	86.61	18
	FS	92.83	5	93.26	5	88.15	5
	FRFS	93.06	3.5	93.41	2.3	90.43	2.3
spambase	F0	87.36	57	90.47	57	79.6	57
	FS	89.49	10.8	90.32	10.8	87.21	10.8
	FRFS	89.44	10.2	90.27	10.4	86.49	9.8
spect-heart	F0	66.15	22	66.41	22	67.31	22
	FS	71.67	3.2	71.03	3.2	70.77	3.2
	FRFS	72.18	2.9	71.03	2.7	71.92	2.6
statlog-heart	F0	81.85	13	74.81	13	83.33	13
	FS	80.37	5.9	77.04	5.9	83.33	5.9
	FRFS	80.62	4.2	77.16	4	82.22	5.3
statlog-image-segmentation	F0	92.19	19	93.32	19	80.66	19
	FS	93.22	5.7	93.36	5.7	82.74	5.7
	FRFS	93.28	4.5	93.45	4.9	84.43	3.9
thoracic-surgery	F0	83.19	16	84.61	16	64.75	16
	FS	84.75	4.4	85.11	4.4	83.83	4.4
	FRFS	84.96	2.7	85.11	2.2	84.4	2.9

classification accuracy of 30 independent experiments using KNN, C4.5 and NB respectively. *F-KNN*, *F-C4.5* and *F-NB* denote the average number of selected features of 30 independent experiments using KNN, C4.5 and NB respectively.

The box plots of performance differences between FRFS and two baselines marked by Post-Friedman test are shown in Figure 1. From the Friedman significance test, we can see that there are significant differences between FRFS and two baselines. FRFS is significantly better than F0 in all learning algorithms and is borderline significantly better than FS in all learning algorithms.

As shown in Table 3, FRFS can achieve smaller number of features than F0 and FS in all learning algorithms. FRFS can achieve better classification accuracy than F0 and FS in C4.5

on 12 out of 15 datasets, can achieve better classification accuracy than F0 and FS in NB on 11 out of 15 datasets, and can achieve better classification accuracy than F0 and FS in KNN on 10 out of 15 datasets.

FS can select several features which have stronger discrimination ability than original features on datasets such as audiology, planning-relax, seismic-bumps, statlog-image-segmentation and thoracic-surgery. However, comparing with FS, FRFS can further reduce the number of features and maintain the classification accuracy, which shows that the feature subset selected by FS is not optimal. For example, on a high-dimensional dataset Lymphoma, FS not only selects about 20 features from original 4026 features but also improves the classification accuracy. However, FRFS further selects about 10 from 20 features selected by FS and improves the classification accuracy. We can get the same results on robot-failure-lp4 dataset and the number of features selected by FRFS has been reduced to nearly 50% comparing with FS.

The experiments in this section confirm our assumption, that is, the more the original features appear in the GP individuals' terminal nodes, the more valuable these features are. Based on this assumption, feature ranking can help feature selection method further reduce the number of features while maintaining the classification accuracy.

5.2. Comparison between FRFS and MFRFS

In this section, MFRFS is compared with FRFS. The experimental results are shown in Table 4. As shown in Table 4, MFRFS can further reduce the number of features comparing with FRFS and can maintain or even improve the classification accuracy in C4.5, KNN and NB learning algorithms on 12, 9 and 8 datasets respectively.

MFRFS further reduces the number of features by more than 50% on arrhythmia dataset. MFRFS can not only select one-third of the number of features but also improve the classification accuracy comparing with FRFS on Lymphoma, especially in NB learning algorithm. As shown in the experimental results, one feature is obtained by MFRFS on climate-simulation-crashes, planning-relax and thoracic-surgery datasets, and is enough to obtain equal or higher classification accuracy than FRFS.

The experimental results show that the multi-criteria fitness function can restrict redundant features as GP's terminal nodes and can achieve the purpose of decreasing the number of selected features while maintaining the classification performance. However, appropriate parameter α in Formula 4 should be set and the effect of parameter α is discussed in Section 5.4.

5.3. Comparison between MFRFS and benchmarks

In this section, MFRFS is compared with five benchmark techniques, i.e. ReliefF, LFS, BGSK, pBGSK and PbGSK-V4. To make a fair comparison with our proposed method, in GSK related methods, the parameter MAXNFE is set to 25,000, the number of generation is set to 500 when the dimension ≤ 20 , and the number of generation is set to 250 when the dimension > 20 . The experimental results are shown in Table 5. The box plots of performance differences between MFRFS and benchmarks are shown in Figure 2.

Table 4. The experimental results of FRFS and MFRFS.

Dataset	Methods	A-KNN	#F-KNN	A-C4.5	#F-C4.5	A-NB	#F-NB
arrhythmia	FRFS	64.42	10.5	66.17	10.5	66.75	11.6
	MFRFS	64.25	4.8	64	4.3	67.25	4.7
audiology	FRFS	70.22	4.9	80	4.9	80	4.9
	MFRFS	72	3.5	81.33	3.3	81.11	3.2
climate-simulation-crashes	FRFS	91.51	2.4	92.2	2	89.12	2.3
	MFRFS	91.89	1	92.2	1	92.45	1
hepatitis	FRFS	82.22	4.6	82	3.3	83.78	5.3
	MFRFS	82	3.2	83.56	2.1	83.56	2.8
house-vote	FRFS	95.74	1	95.63	1	95.63	1
	MFRFS	95.74	1	95.63	1	95.63	1
Lymphoma	FRFS	80	11.1	90	9.8	84	10.8
	MFRFS	79.33	3.6	92	3.3	92.67	3.3
planning-relax	FRFS	68.52	2.7	70.93	2.1	68.89	2.4
	MFRFS	67.04	1	71.67	1	71.67	1
robot-failure-lp4	FRFS	61.82	11.7	76.97	11.9	83.03	16.2
	MFRFS	66.06	3.9	78.18	5	77.27	6
robot-failure-lp5	FRFS	46.9	15.4	46.43	19.7	62.86	21.8
	MFRFS	53.33	4.3	46.9	5.5	60.48	5.3
seismic-bumps	FRFS	93.06	3.5	93.41	2.3	90.43	2.3
	MFRFS	93.32	1.1	93.41	1	91.82	1.1
spambase	FRFS	89.44	10.2	90.27	10.4	86.49	9.8
	MFRFS	89.33	6.5	89.43	6.4	84.61	5.8
spect-heart	FRFS	72.18	2.9	71.03	2.7	71.92	2.6
	MFRFS	72.31	1.9	68.72	1.8	71.41	1.8
statlog-heart	FRFS	80.62	4.2	77.16	4	82.22	5.3
	MFRFS	80.99	3.2	78.4	3.2	81.11	4.3
statlog-image-segmentation	FRFS	93.28	4.5	93.45	4.9	84.43	3.9
	MFRFS	93.85	3	93.78	3	84.39	3
thoracic-surgery	FRFS	84.96	2.7	85.11	2.2	84.4	2.9
	MFRFS	84.75	1	85.11	1	85.04	1

5.3.1. Comparison between MFRFS and ReliefF

Like MFRFS, ReliefF also needs to rank the features. In the experiment of ReliefF, we select the same number of features as MFRFS to verify whether MFRFS can achieve higher classification accuracy when the two methods take the same number of features.

On statlog-image-segmentation dataset, both MFRFS and ReliefF select three features, but the accuracy of MFRFS is 10% higher than that of ReliefF. MFRFS and ReliefF can achieve comparable classification accuracy with only one feature on climate-simulation-crashes, seismic-bumps, thoracic-surgery and planning-relax datasets. On arrhythmia, hepatitis, robot-failure-lp4, robot-failure-lp5, spambase and statlog-heart datasets, MFRFS can achieve better classification accuracy with same number of features than ReliefF. From the experimental results in Table 5, we can see that MFRFS is more robust than ReliefF. The box plots marked by Post-Friedman test in Figure 2 show that MFRFS is significantly better than ReliefF in KNN and NB learning algorithms, and is borderline significantly better than ReliefF in C4.5 learning algorithm. In general, the experimental results show that MFRFS can obtain better classification results than ReliefF.

5.3.2. Comparison between MFRFS and LFS

The comparison results between MFRFS and LFS are shown in Table 5. As shown in Table 5, in general, LFS can achieve comparable or better classification accuracy than MFRFS. The

Table 5. Experimental results of different feature selection methods.

Dataset	Methods	A-KNN	#F-KNN	A-C4.5	#F-C4.5	A-NB	#F-NB
arrhythmia	MFRFS	64.25	4.8	64	4.3	67.25	4.7
	ReliefF	59.83	4.8	62.08	4.3	59.75	4.7
	LFS	64.58	18.8	68.25	18.8	66.42	18.8
	BGSK	61.64	15.9	59.72	15.9	56.61	15.9
	pBGSK	60.5	20	58.83	20	54.53	20
audiology	PbGSK-V4	61.89	1.2	61.72	1.2	63.06	1.2
	MFRFS	72	3.5	81.33	3.3	81.11	3.2
	ReliefF	75.78	3.5	80	3.3	79.11	3.2
	LFS	67.33	15.3	79.33	15.3	76.89	15.3
	BGSK	67.93	5.5	80.96	5.5	77.04	5.5
climate-simulation-crashes	pBGSK	67.26	7	80.67	7	75.93	7
	PbGSK-V4	39.56	1	39.56	1	39.56	1
	MFRFS	91.89	1	92.2	1	92.45	1
	ReliefF	91.76	1	92.2	1	92.45	1
	LFS	91.19	2.9	91.76	2.9	86.86	2.9
hepatitis	BGSK	91.51	2.1	92.26	2.1	86.42	2.1
	pBGSK	91.4	1.9	92.24	1.9	87.86	1.9
	PbGSK-V4	92.05	1	92.43	1	92.45	1
	MFRFS	82	3.2	83.56	2.1	83.56	2.8
	ReliefF	80.67	3.2	79.11	2.1	78.89	2.8
house-vote	LFS	83.78	7.4	82	7.4	84	7.4
	BGSK	80.37	4.2	80.22	4.2	83.48	4.2
	pBGSK	80.74	4.1	80.22	4.1	83.56	4.1
	PbGSK-V4	80	1.4	80.52	1.4	80.3	1.4
	MFRFS	95.74	1	95.63	1	95.63	1
Lymphoma	ReliefF	95.74	1	95.63	1	95.63	1
	LFS	94.47	3	95.61	3	94.92	3
	BGSK	95.34	1.5	95.63	1.5	95.21	1.5
	pBGSK	95.45	1.3	95.53	1.3	95.4	1.3
	PbGSK-V4	94.74	2.2	95.5	2.2	94.89	2.2
planning-relax	MFRFS	79.33	3.6	92	3.3	92.67	3.3
	ReliefF	78	3.6	92.67	3.3	92.67	3.3
	LFS	80	73.1	91.33	73.1	80	73.1
	BGSK	80	1319.9	84.44	1319.9	80	1319.9
	pBGSK	80	1395.5	85.11	1395.5	80	1395.5
robot-failure-lp4	PbGSK-V4	78	1.2	80.89	1.2	77.33	1.2
	MFRFS	67.04	1	71.67	1	71.67	1
	ReliefF	65.39	1	71.48	1	70.56	1
	LFS	63.52	1	72.22	1	72.04	1
	BGSK	66.36	1	71.98	1	71.05	1
robot-failure-lp5	pBGSK	66.36	1	72.16	1	71.48	1
	PbGSK-V4	68.09	1	72.16	1	71.17	1
	MFRFS	66.06	3.9	78.18	5	77.27	6
	ReliefF	75.76	3.9	71.21	5	48.48	6
	LFS	57.58	29.3	75.15	29.3	81.52	29.3
seismic-bumps	BGSK	65.05	8.1	71.01	8.1	76.77	8.1
	pBGSK	63.13	9.4	68.89	9.4	76.77	9.4
	PbGSK-V4	68.18	1.1	67.17	1.1	66.26	1.1
	MFRFS	53.33	4.3	46.9	5.5	60.48	5.3
	ReliefF	46.9	4.3	42.62	5.5	39.05	5.3
	LFS	43.81	23.4	46.43	23.4	58.33	23.4
	BGSK	49.68	7.9	49.44	7.9	60.87	7.9
	pBGSK	48.97	10.3	47.46	10.3	60.08	10.3
	PbGSK-V4	38.41	1.2	39.92	1.2	35.32	1.2
	MFRFS	93.32	1.1	93.41	1	91.82	1.1
	ReliefF	93.41	1.1	93.41	1	93.14	1.1
	LFS	92.79	5.2	93.26	5.2	88.14	5.2
	BGSK	92.86	1.8	93.35	1.8	90.7	1.8
	pBGSK	92.89	1.9	93.35	1.9	90.53	1.9

(continued)

Table 5. Continued.

Dataset	Methods	A-KNN	#F-KNN	A-C4.5	#F-C4.5	A-NB	#F-NB
spambase	PbGSK-V4	93.16	1.4	93.41	1.4	92.76	1.4
	MFRFS	89.33	6.5	89.43	6.4	84.61	5.8
	ReliefF	80.3	6.5	82.42	6.4	67.46	5.8
	LFS	90.07	16.7	90.81	16.7	80.68	16.7
	BGSK	89.3	7.2	89.41	7.2	82.08	7.2
	pBGSK	89.08	8.1	89.38	8.1	83.95	8.1
	PbGSK-V4	74.44	1	74.67	1	70.19	1
	MFRFS	72.31	1.9	68.72	1.8	71.41	1.8
spect-heart	ReliefF	71.03	1.9	68.97	1.8	70.64	1.8
	LFS	71.41	3.6	70.9	3.6	70.64	3.6
	BGSK	73.21	2.3	71.32	2.3	71.75	2.3
	pBGSK	72.91	2.5	71.32	2.5	71.92	2.5
	PbGSK-V4	71.32	1	71.11	1	71.28	1
	MFRFS	80.99	3.2	78.4	3.2	81.11	4.3
	ReliefF	73.09	3.2	75.56	3.2	78.64	4.3
	LFS	79.26	6.7	76.91	6.7	82.96	6.7
statlog-heart	BGSK	79.96	4.3	78.48	4.3	80.82	4.3
	pBGSK	80.12	4.3	78.48	4.3	80.82	4.3
	PbGSK-V4	63.87	1.5	63.62	1.5	63.79	1.5
	MFRFS	93.85	3	93.78	3	84.39	3
	ReliefF	82.6	3	83.2	3	73.51	3
	LFS	92.51	7.6	93.9	7.6	84.24	7.6
	BGSK	93.51	3.1	94	3.1	84.05	3.1
	pBGSK	93.72	3	94.01	3	84.08	3
statlog-image-segmentation	PbGSK-V4	74.56	1.6	75.04	1.6	65.65	1.6
	MFRFS	84.75	1	85.11	1	85.04	1
	ReliefF	84.4	1	85.11	1	84.61	1
	LFS	84.68	4.9	85.11	4.9	83.97	4.9
	BGSK	85.06	1.3	85.11	1.3	85.06	1.3
	pBGSK	85.11	1.3	85.11	1.3	85.01	1.3
	PbGSK-V4	85.11	1	85.11	1	85.15	1

box plots marked by Post-Friedman test in Figure 2 show that MFRFS is borderline significantly better than LFS in KNN and NB learning algorithms, however, LFS selects more features than MFRFS.

On arrhythmia and audiology datasets, the number of features selected by LFS is nearly four times that of MFRFS. However, on audiology dataset, LFS does not obtain a better classification accuracy with more features than MFRFS. On robot-failure-lp4 and robot-failure-lp5 datasets, features selected by LFS are nearly six times that of MFRFS. On the Lymphoma dataset, LFS even uses more than 20 times features that of MFRFS, and when NB is used as the learning algorithm, the accuracy of LFS is 10% less than that of MFRFS. On other datasets, MFRFS can select smaller number of features and achieve comparable or better classification accuracy than LFS. The experimental results show that LFS's dimensionality reduction performance is weak. The features selected by LFS still contains irrelevant and redundant features, which confirm that some invalid features are not helpful for classification. Our proposed MFRFS can select more effective features on the premise of maintaining the classification accuracy.

5.3.3. Comparison between MFRFS and GSK related methods (BGSK, pBGSK, PbGSK-V₄)

Compared with BGSK, MFRFS selects a smaller feature subset and achieves better classification performance on most cases in all learning algorithms. On the higher-dimensional

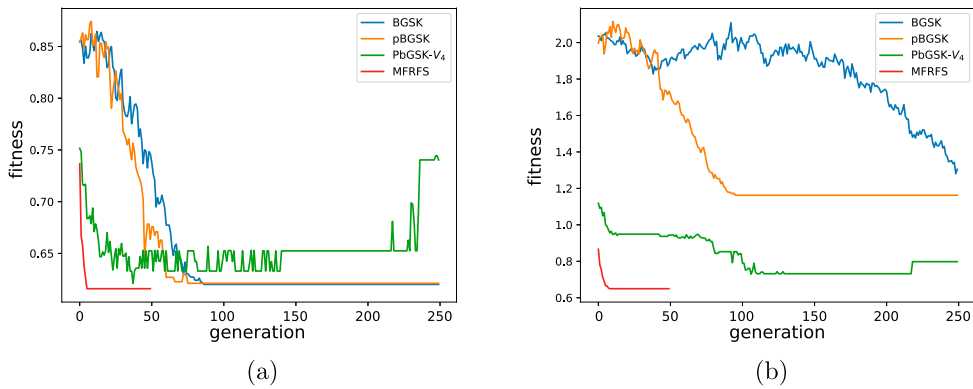


Figure 3. Convergence graph on two datasets: (a) spambase and (b) arrhythmia.

dataset Lymphoma, BGSK cannot reduce the dimension of features better, and select more than 1300 features. We obtain similar experimental results on pBGSK, and in most cases, a larger feature subset is selected, especially on the Lymphoma dataset. The box plots marked by Post-Friedman test in Figure 2 show that MFRFS is borderline significantly better than pBGSK in KNN learning algorithm and significantly better than pBGSK in NB learning algorithm.

As shown in Table 5, PbGSK-V₄ selects minimum number of features among the six methods. However, this method cannot achieve good classification performance, especially on audiology, robot-failure-lp5, statlog-image-segmentation and spambase datasets. Only an average of less than 2 features is selected by PbGSK-V₄ in those datasets, but the classification accuracy is very low. Compared with MFRFS, PbGSK-V₄ reduces the number of features, but in most cases the overall classification accuracy of MFRFS is better than PbGSK-V₄.

Figure 3 shows the convergence behaviours of four EC-based methods on arrhythmia and spambase datasets. As shown in Figure 3, MFRFS starts to converge in about 10 generations and can obtain good solutions. The GSK related methods need more generations to converge.

5.4. The parameter setting of α in MFRFS

The parameter α in Formula 4 is used to restrict redundant and irrelevant features as GP's terminal nodes and further reduce the number of selected features. In Section 5.2, we have shown that MFRFS can further reduce the number of features comparing with FRFS. To show the impact of parameter α on experimental results and set an appropriate parameter, different values of parameter α are experimented and the experimental results are shown in Table 6. As shown in Table 6, the number of features selected by MFRFS is inversely proportional to the value of α , which means that the smaller the value of α , the more the number of features selected.

When $\alpha = 0.001$, the classification performance of MFRFS is similar to FRFS, and the parameter α has little effect on classification performance. When $\alpha = 0.005$, it has a certain degree of dimensionality reduction effect. When $\alpha = 0.01$, it can be seen from Table 6 that the overall performance is the best. When $\alpha = 0.025$, the number of features further

Table 6. Experimental results of different α values.

Dataset	α	A-KNN	#F-KNN	A-C4.5	#F-C4.5	A-NB	#F-NB
arrhythmia	0.1	59.17	1	60.92	1	61.83	1
	0.05	60.33	1.7	62.5	1.7	61.42	1.7
	0.025	62.08	2.9	62.17	2.7	65	3.1
	0.01	64.25	4.8	64	4.3	67.25	4.7
	0.005	63.83	5.9	61.25	6.1	65.58	6.4
audiology	0.001	64.58	9.7	66.67	9.3	64.83	11.3
	0.1	60	2	60	2	60	2
	0.05	60	2	60	2	60	2
	0.025	80.22	2.9	80.22	2.9	79.56	2.9
	0.01	72	3.5	81.33	3.3	81.11	3.2
climate-simulation-craches	0.005	64.89	4	80.44	4.2	79.33	4
	0.001	69.11	5.3	82.44	4.9	78.67	5.1
	0.1	91.76	1	92.2	1	92.45	1
	0.05	91.7	1	92.2	1	92.45	1
	0.025	91.7	1	92.2	1	92.45	1
hepatitis	0.01	91.89	1	92.2	1	92.45	1
	0.005	91.38	2	92.2	1.6	91.45	1.6
	0.001	91.26	2.8	92.2	2	89.31	2.4
	0.1	79.56	1	80.44	1	80	1
	0.05	80.89	1.2	80	1.5	80.44	1.2
house-vote	0.025	79.56	2.5	79.78	2.1	81.78	1.8
	0.01	82	3.2	83.56	2.1	83.56	2.8
	0.005	79.78	3	79.11	2.4	81.33	3.6
	0.001	81.78	4.5	81.56	2.8	82.89	5
	0.1	95.74	1	95.63	1	95.63	1
Lymphoma	0.05	95.74	1	95.63	1	95.63	1
	0.025	95.74	1	95.63	1	95.63	1
	0.01	95.74	1	95.63	1	95.63	1
	0.005	95.74	1	95.63	1	95.63	1
	0.001	95.74	1	95.63	1	95.63	1
planning-relax	0.1	79.33	1.2	89.33	1.2	91.33	1.2
	0.05	77.33	1.4	90.67	1.4	90.67	1.5
	0.025	79.33	2.8	90.67	2.2	86	2.7
	0.01	79.33	3.6	92	3.3	92.67	3.3
	0.005	79.33	4.3	92	4.2	87.33	4.1
robot-failure-lp4	0.001	80	6.5	87.33	6.7	88.67	6.1
	0.1	65.19	1	70.93	1	71.3	1
	0.05	65.93	1	71.85	1	71.48	1
	0.025	65	1	71.85	1	71.11	1
	0.01	67.04	1	71.67	1	71.67	1
robot-failure-lp5	0.005	66.67	1	71.85	1	70.93	1
	0.001	64.63	1	71.85	1	71.11	1
	0.1	73.03	1.5	69.7	1.3	66.06	1.4
	0.05	73.33	2.3	73.33	2.2	69.7	2.1
	0.025	70.3	3.2	75.45	3	73.64	3.7
robot-failure-lp4	0.01	66.06	3.9	78.18	5	77.27	6
	0.005	68.79	4.8	77.27	4.8	78.48	6.9
	0.001	67.58	9.2	76.97	9.6	78.18	15
	0.1	40.95	1	40.24	1	40.48	1
	0.05	51.9	2.4	50.24	2.3	53.1	2.4
robot-failure-lp5	0.025	50.95	3.2	45.48	3.3	58.57	3.7
	0.01	53.33	4.3	46.9	5.5	60.48	5.3
	0.005	47.62	7.3	43.1	8.9	57.86	8.2
	0.001	46.9	10.2	46.9	13.9	57.86	13.7
	0.1	93.27	1	93.41	1	91.95	1
robot-failure-lp5	0.05	93.27	1	93.41	1	91.95	1

(continued)

Table 6. Continued.

Dataset	α	A-KNN	#F-KNN	A-C4.5	#F-C4.5	A-NB	#F-NB
seismic-bumps	0.025	93.27	1	93.41	1	91.95	1
	0.01	93.32	1.1	93.41	1	91.82	1.1
	0.005	93.06	1.7	93.37	1.4	91.02	1.5
	0.001	93.09	2.8	93.41	2.1	90.4	2.1
	0.1	76.31	1	76.44	1	72.19	1
	0.05	84.52	2	85.04	2	78.39	2
spambase	0.025	87.72	3.97	87.8	3.9	80.07	3.83
	0.01	89.33	6.53	89.43	6.43	87.4	5.83
	0.005	89.27	8	89.51	7.97	85.36	7.43
	0.001	89.57	9.37	89.99	9.43	84.61	9
	0.1	69.62	1	69.62	1	69.62	1
	0.05	69.62	1	69.62	1	69.62	1
spect-heart	0.025	71.92	1.5	70.38	1.4	71.41	1.5
	0.01	72.31	1.9	68.72	1.8	71.41	1.8
	0.005	72.56	1.9	69.49	1.8	71.28	1.8
	0.001	72.31	2.4	69.87	2	71.15	2.1
	0.1	71.73	1	70.99	1	71.11	1
	0.05	71.73	1.5	72.35	1.1	74.2	1.8
statlog-heart	0.025	77.41	2.8	77.16	2.6	77.16	3
	0.01	80.99	3.2	78.4	3.2	81.11	4.3
	0.005	79.75	4.2	77.53	4.3	81.36	4.9
	0.001	78.64	3.8	76.67	4.3	81.98	5.3
	0.1	82.24	1.9	82.34	1.9	72.21	1.9
	0.05	84.49	2	84.47	2	73.59	2
statlog-image-segmentation	0.025	89.13	2.5	89.08	2.5	78.96	2.5
	0.01	93.85	3	93.78	3	84.39	3
	0.005	93.85	3	93.78	3.4	84.43	3.1
	0.001	93.75	3.9	93.49	4.1	84.5	3.7
	0.1	84.75	1	85.11	1	85.04	1
	0.05	84.75	1	85.11	1	85.04	1
thoracic-surgery	0.025	84.75	1	85.11	1	85.04	1
	0.01	84.75	1	85.11	1	85.04	1
	0.005	85.11	1.5	85.11	1.2	84.89	1.6
	0.001	84.96	2.6	85.11	1.9	84.26	2.5

decreases, but the average classification accuracy starts to decline to some extent. When $\alpha = 0.5$ and 0.1, the number of features is obviously reduced a lot, but the accuracy at this time is also declined a lot.

Therefore, the parameter α has a great impact on the classification accuracy. If the parameter α is too small, it cannot achieve the purpose of dimensionality reduction comparing with FRFS. If the parameter α is too large, it will remove effective features and reduce the classification accuracy. Through the experiments, $\alpha = 0.01$ meets the purpose of decreasing the number of selected features while maintaining the classification performance.

5.5. Further discussion

In this paper, FRFS, a feature selection method combined with feature ranking using GP, is proposed to select smaller number of features comparing with feature selection method only (FS). Based on a multi-criteria fitness function, MFRFS is proposed to obtain smaller number of features comparing with FRFS. Here are some further discussions.

(1) Evolutionary computation algorithms will produce a large number of individuals with the best fitness. If only one of the best individuals is used for feature selection, some good

Table 7. Classification accuracy of different number of top features.

f_{best}	Training set			Testing set		
	acc_{KNN}	$acc_{\text{C4.5}}$	acc_{NB}	acc_{KNN}	$acc_{\text{C4.5}}$	acc_{NB}
12	62.35	68.98	70.19	65.83	61.67	61.67
11	63.56	69.58	67.47	66.67	64.17	65.00
10	64.16	71.69	68.37	65.83	68.33	57.50
9	68.07	71.39	70.18	63.33	66.67	66.67
8	66.27	65.66	67.17	59.17	60.83	62.5
7	65.96	65.96	67.77	60.00	60.00	65.00
6	66.27	68.37	68.07	63.33	60.83	65.00
5	65.06	66.57	66.57	62.50	58.33	65.83

individuals may be lost. In this paper, GP individuals after the 30th generation with the best fitness are preserved. The features are ranked according to their occurrence times in these individuals. Experiments show that feature ranking can help select more effective features.

To further demonstrate how feature ranking works, we take the arrhythmia dataset as an example. The results of feature ranking are shown below. A total of 12 best features were selected. The values before the “-” represent the index of the feature, and the values after the “-” represent the number of times the feature occurs in all the best individuals.

F14-60244 F250-59799 F109-58235 F182-42527 F212-38822 F195-38505

F241-37687 F7-31955 F208-26438 F38-24976 F97-23970 F10-21086

We can see that feature F14 is probably the best feature, and feature F250 is the second best, and so on. The purpose of feature ranking is to select the best feature set from the top 12 features.

According to the method described in Section 3.2, the classification accuracy on different number of top ranked features on training set and testing set in f_{best} feature subset are shown in Table 7. For training set, we can see that the classification accuracy first increases and then decreases with the increase of the number of top features. When the number of top features is 9, 10 and 9 for KNN, C4.5 and NB, the classification accuracy is the highest respectively.

For the testing set, we also obtain the classification accuracies of the corresponding ranking features for KNN, C4.5 and NB learning algorithms respectively in Table 7, and feature ranking can achieve highest classification accuracy on corresponding ranking features in C4.5 and NB learning algorithms.

(2) Feature ranking is based on the assumption that the more the original features appear in the GP individuals’ terminal nodes, the more valuable these features are. Experiments prove that the assumption is correct. GP can be used to quantify the importance of features and select more important feature sets.

(3) MFRFS can restrict the number of irrelevant and redundant features in the GP’s terminal nodes by a multi-criteria fitness function. By adjusting the parameter α , MFRFS can achieve a smaller number of features while maintaining the classification accuracy compared with FRFS.

(4) In general, MFRFS can achieve better classification performance than the five benchmarks. However, MFRFS needs to store the individuals with the best fitness, rank the features according to the occurrence times of features appearing in the GP individuals’ terminal nodes and search smaller number of features with higher classification performance from the ranked features. Therefore MFRFS has no advantages in time consumption and

computational complexity, especially on datasets with more features. In practical applications, we need to find a balance between classification performance and computational complexity.

(5) Feature selection is a key data preprocessing process, which can solve “the curse of dimensionality” problems. Our proposed MFRFS can greatly reduce the dimension of features and obtain a better classification performance. The most important thing is that we can identify which features are more important through the feature ranking method in this paper. In industrial applications, our methods may greatly reduce the model’s training time and improve the classification performance.

6. Conclusion and future work

This paper proposes a feature selection method with feature ranking (FRFS) using GP, which ranks the features according to the number of their occurrence in the individuals with the best fitness. Based on FRFS, to further reduce the number of selected features, another feature selection method using a multi-criteria fitness function (MFRFS) is proposed. Experiments on 15 datasets show that FRFS can achieve better classification performance and obtain smaller number of features than FS. Compared with FRFS, MFRFS can further remove redundant and irrelevant features, and select smaller number of features while maintaining the classification performance. Compared with other five existing feature selection methods, MFRFS can achieve better classification performance.

Feature selection is important to build an efficient classification model, and feature ranking has been proved to be effective for feature selection. The fitness function of our methods in this paper are based on Correlation. It is necessary to further investigate whether our methods are effective to other evaluation criteria. Moreover, the dimension of the selected datasets is relatively low. In the future, feature ranking methods on high dimensional datasets will be investigated.

Disclosure statement

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